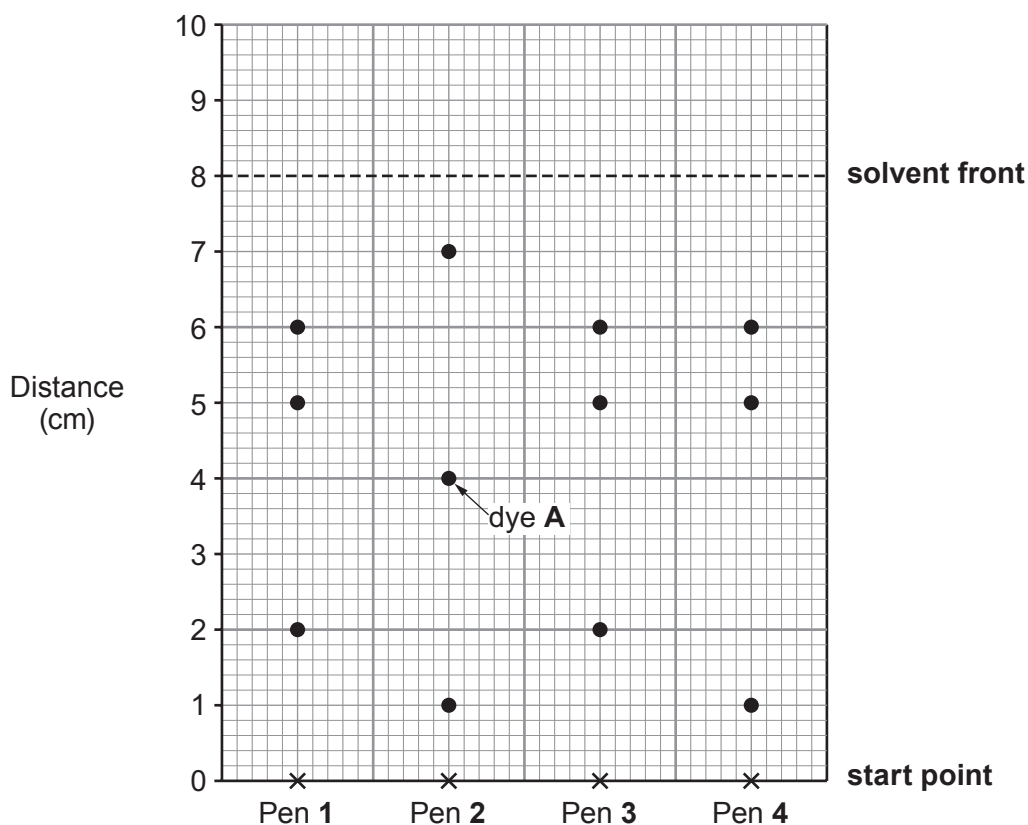


- (b) A group of students were asked to investigate the dyes in four different water soluble blue pens, **1**, **2**, **3** and **4**. The results are shown below.



- (i) Which **two** pens contain the same three dyes? [1]
Pens and
- (ii) Which pen contains the **most** soluble dye? [1]
Pen
- (iii) The R_f value of a substance can be used to identify that substance.
The R_f value is given by the formula:

$$R_f = \frac{\text{distance moved by the substance}}{\text{distance moved by the solvent front}}$$

Calculate the R_f value for dye **A**. [2]

$R_f =$



(iv) The box contains methods for separating mixtures.

distillation	chromatography	filtration
---------------------	-----------------------	-------------------

Complete the sentence using a method from the box.

The method used to separate the coloured dyes in the blue pens is called

.....

Examiner
only

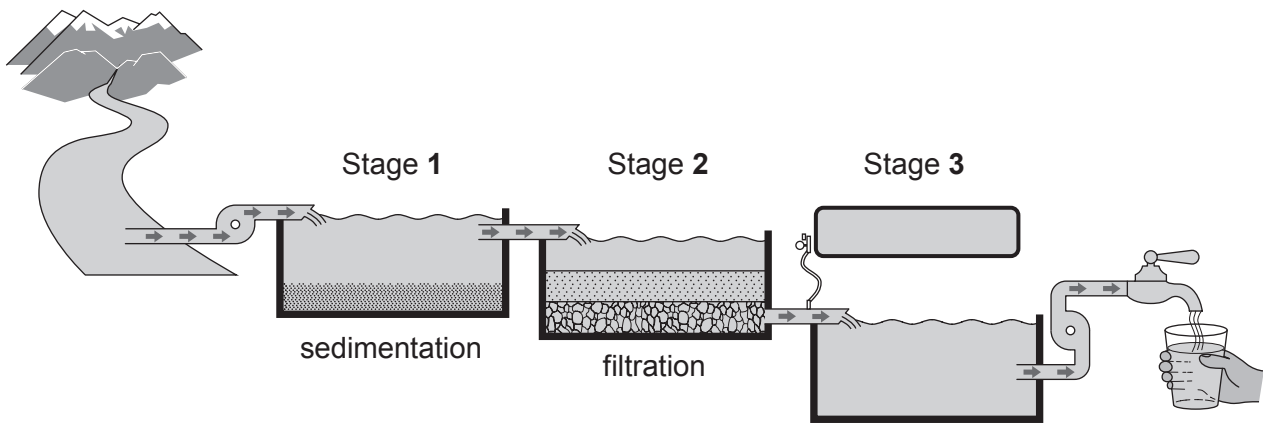
[1]

8

3410U101
05

4. (a) Most water in Wales is sourced from surface water and ground water. Ground water is the water beneath the surface of the Earth, consisting mainly of surface water that has filtered through the rocks. It is the source of water found in springs and wells. Human activity causes surface water to be more contaminated than ground water.

The diagram below shows the three main stages in the treatment of surface water before it enters the mains water system on its way to the taps in your home.



- (i) Tick (✓) the box that describes what occurs in stage 1. [1]

- | | |
|--|--------------------------|
| large insoluble particles sink to the bottom | ☐ |
| soluble particles are removed | ☐ |
| small fine insoluble particles are removed | ☐ |
| fluoride is added to the water | ☐ |

- (ii) Stage 3 makes water safe to drink. Name the substance added to water in stage 3. State why this makes the water safe to drink. [2]

Substance

.....

- (iii) Describe **one major** cause of surface water pollution. [2]

.....

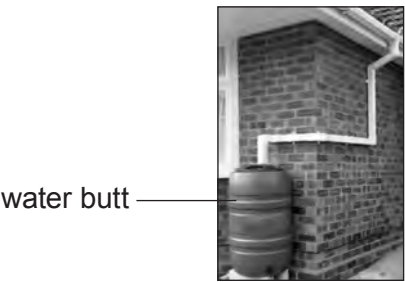
.....

.....

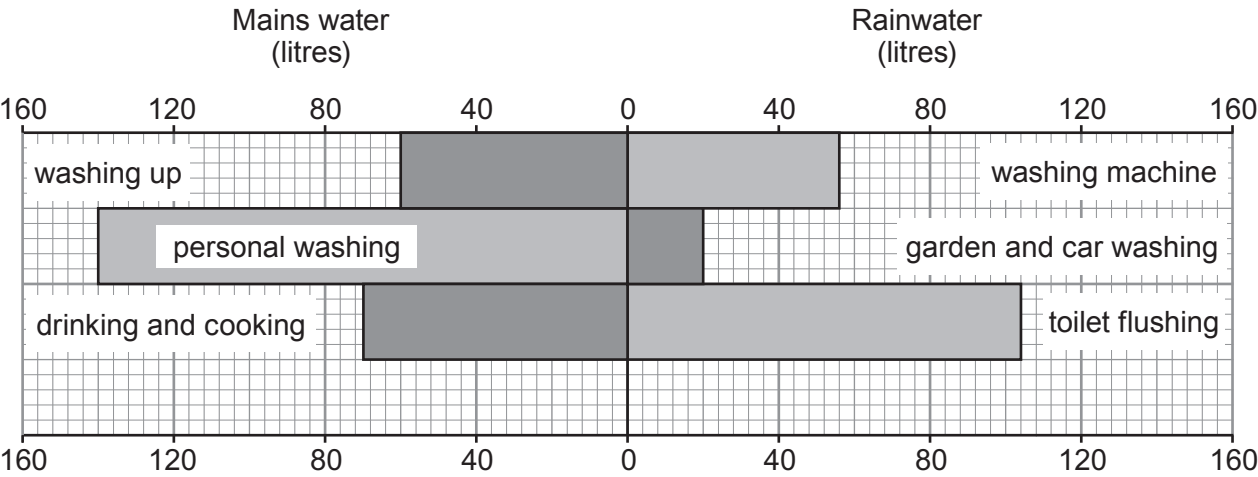
.....

Examiner
only

(b) A family uses a water butt to collect rainwater in order to reduce their use of mains water.



They use a total of 450 litres of water on a given day. The chart shows the volumes of mains water and collected rainwater used for various purposes on that day.



(i) What percentage of the total water used on that day is rainwater? [2]

percentage rainwater = %

(ii) Suggest why it is safe to use rainwater for the uses shown on the right hand side of the chart. [1]

.....

.....

4. The following table shows the composition of the atmosphere.

Gas	Chemical formula	Percentage found in the atmosphere (%)
argon	Ar	0.93
carbon dioxide	CO ₂	0.0360
helium	He	0.0005
hydrogen	H ₂	0.00005
methane	CH ₄	0.00017
neon	Ne	0.0018
nitrogen	N ₂	78.08
nitrous oxide	N ₂ O	0.00003
oxygen	O ₂	20.95
ozone	O ₃	0.000004

Use the table to answer parts (a) and (b).

(a) (i) Name **two** gases that occur as single atoms. [1]

.....

(ii) Name **two** elements that occur as molecules. [1]

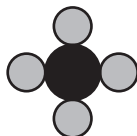
.....

(iii) Name the gas that has the **lowest** percentage. [1]

.....



- (b) (i) If  represents a carbon atom, name the gases from the table that can be represented by the following diagrams. [2]



.....

- (ii) Use the information in part (i) to draw the diagram for oxygen gas. [1]

- (c) The atmosphere can become polluted by substances known as chlorofluorocarbons (CFCs). One example of a CFC is CH_2ClF .

- (i) State how many hydrogen atoms are present in CH_2ClF . [1]

.....

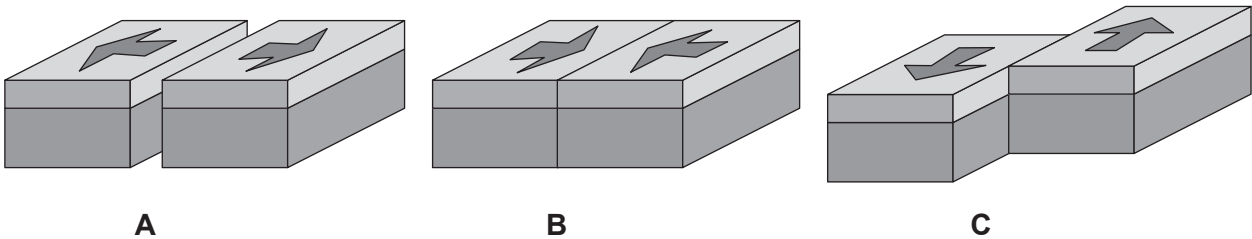
- (ii) Give the **total** number of atoms shown in the formula, CH_2ClF . [1]

.....



4. The surface of the Earth is divided into a number of tectonic plates. These plates are constantly moving.

Plate boundaries can be classified according to the direction of movement of the plates. The three main types of boundary are shown below.



conservative	destructive	constructive
---------------------	--------------------	---------------------

(a) Choose the name of each type of boundary from the box. [2]

A
B
C

(b) Describe what happens as a result of the movement shown at boundary **A**. [2]

.....
.....
.....

(c) Explain what happens at boundary **C** as a result of the plate movement shown. [2]

.....
.....
.....

6



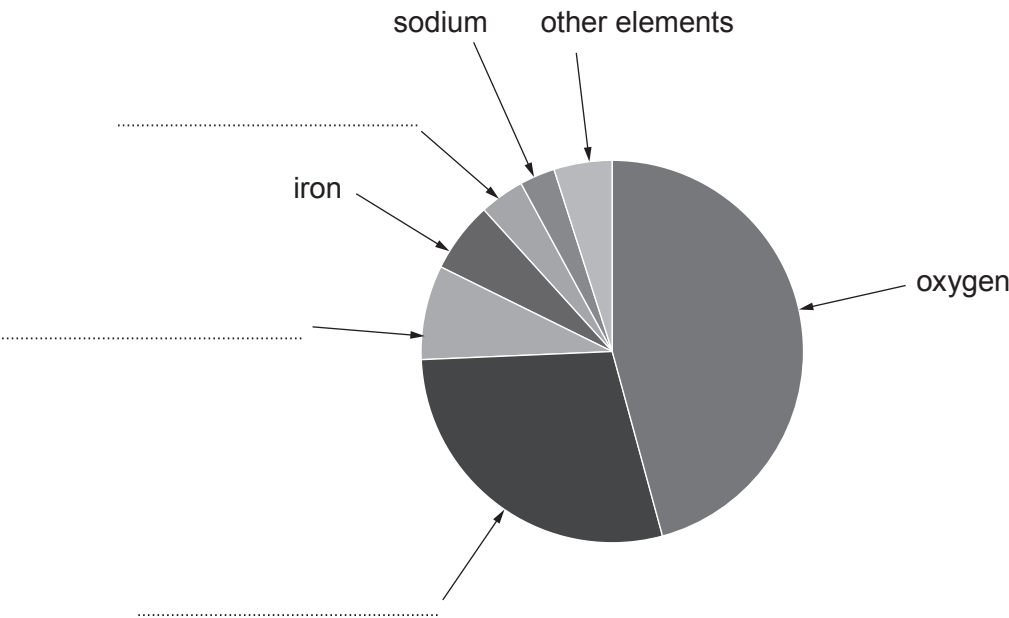
4. (a) The Earth’s crust contains more than 80 elements, including many metals. Some of the metals, such as gold, are found ‘native’ or uncombined, but most are bonded with other elements to form compounds.

The crust can be thicker than 80 kilometres in some spots and less than one kilometre thick in others.

Just six elements make up almost all of the Earth’s crust. These are aluminium, calcium, iron, oxygen, silicon and sodium.

Silicon is the second most abundant element. The amount of aluminium is approximately double the amount of calcium.

The chart shows the percentages of elements contained in the crust.



- (i) Use the information given to **label** the remaining sections of the chart. [2]
- (ii) Underline the approximate fraction of elements in the crust which are metals. [1]

$\frac{1}{2}$ $\frac{1}{3}$ $\frac{1}{4}$ $\frac{3}{4}$



(iii) Tick (✓) the statement which best explains why most metals are **not** found native in the crust. [1]

- most metals have higher melting points than gold

☐
- most metals are magnetic

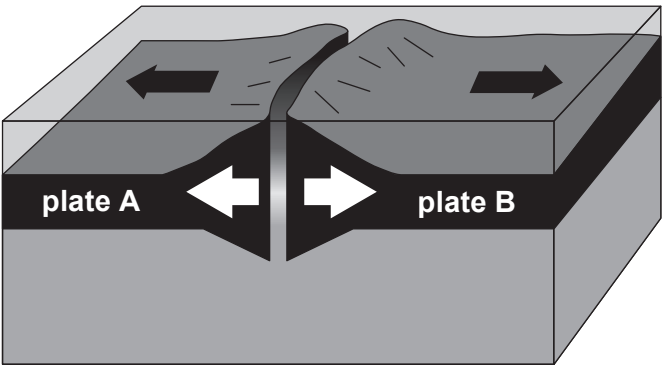
☐
- most metals are more reactive than gold

☐
- most metals are radioactive

☐

(b) Tectonic plates are huge sections of the crust that move very slowly above the mantle. The places where plates meet are called plate boundaries.

The diagram shows one type of plate boundary.



(i) Describe how new rock is formed at this plate boundary. [3]

(ii) Underline the name of this type of plate boundary. [1]

conservative

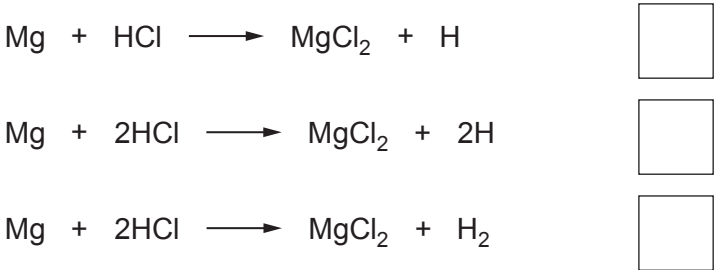
destructive

constructive

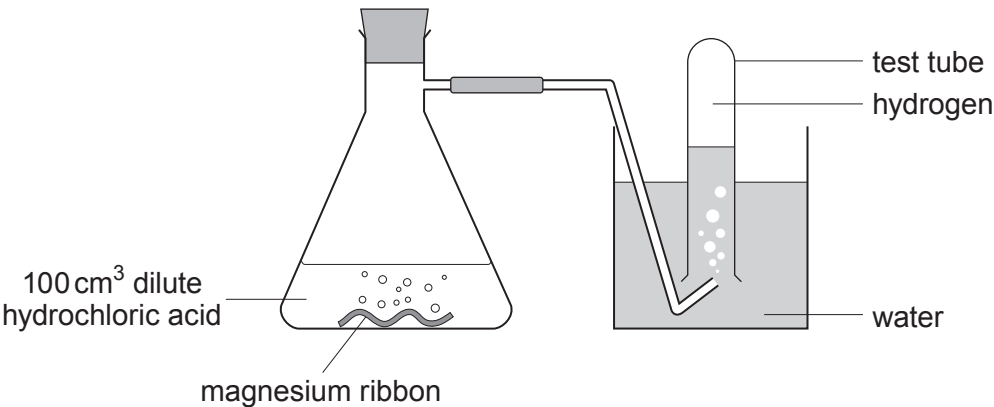


4. Dilute hydrochloric acid reacts with magnesium forming magnesium chloride and hydrogen gas.

(a) Tick (✓) the box next to the correct equation for the reaction between magnesium and hydrochloric acid. [1]



(b) Osian wanted to find out how changing the concentration of the acid affects the rate of the reaction. He carried out five experiments at room temperature (20 °C). He added a 4 cm piece of magnesium ribbon to 100 cm³ of hydrochloric acid of five different concentrations. He recorded the time it took to half-fill a test tube with gas.

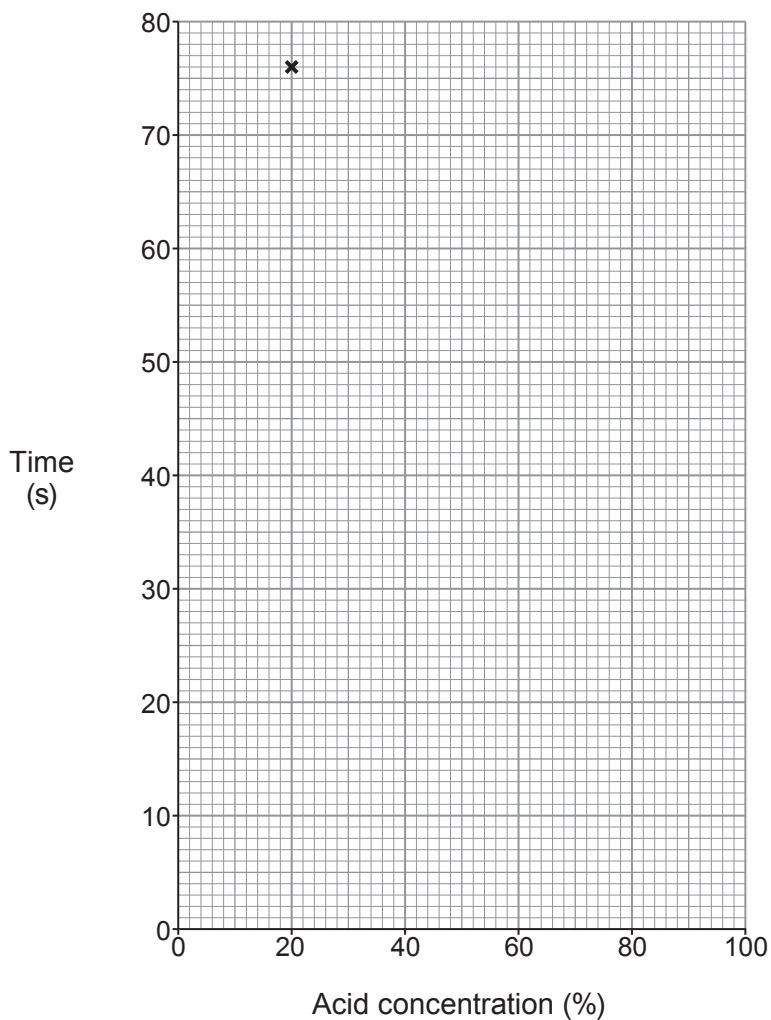


His results are shown below.

Experiment	Acid concentration (%)	Time (s)
1	100	12
2	80	14
3	60	20
4	40	36
5	20	76



- (i) Plot the acid concentration against time on the grid below and draw a suitable line. One point has been plotted for you. [3]



- (ii) Underline the correct word(s) in the brackets to complete the following sentences. [2]

As the acid concentration increases, the **time** to half-fill the test tube with gas
(**increases** / **stays the same** / **decreases**).

As the acid concentration increases, the **rate** of the reaction
(**increases** / **stays the same** / **decreases**).



- (iii) Using your knowledge of particle theory, underline the correct words in the brackets to complete the following sentence. [2]

At a higher concentration, there are (**more** / **less** / **the same number of**) particles present so there will be (**an equal** / **a smaller** / **a greater**) chance of collision.

- (iv) There are other ways the rate of the reaction can be changed.

Tick (✓) the **two** statements that correctly describe other ways the rate of reaction can be increased. [2]

Increasing the temperature of the acid

☐

Using a lump of magnesium

☐

Using a different apparatus

☐

Using magnesium powder

☐

Decreasing the temperature of the acid

☐